Policy Gradients

Go kaboom

Rushil Gupta & Dhruman Gupta

The table is getting BIG

Q-learning stores a value $Q(s, a)$ for every state and action

	a_1	a_2	a_3	a_4	⋯
s_1	0.3	1.2	−0.4	0.8	⋯
s_2	−0.1	0.5	0.9	0.0	⋯
s_3	1.4	−0.7	0.2	0.6	⋯
s_4	0.1	0.3	−0.2	1.1	⋯
s_5	−0.5	0.7	0.4	−0.3	⋯
⋮	⋮	⋮	⋮	⋮	⋯

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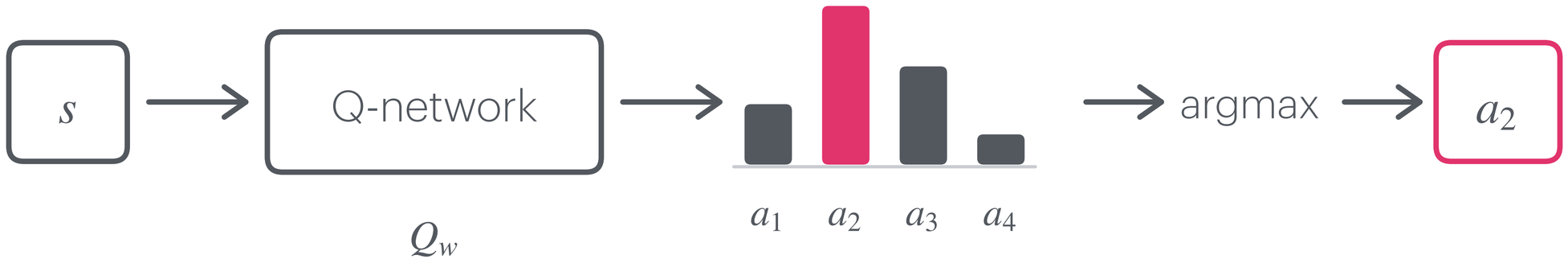
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$|\mathcal{S}| \times |\mathcal{A}|$ entries

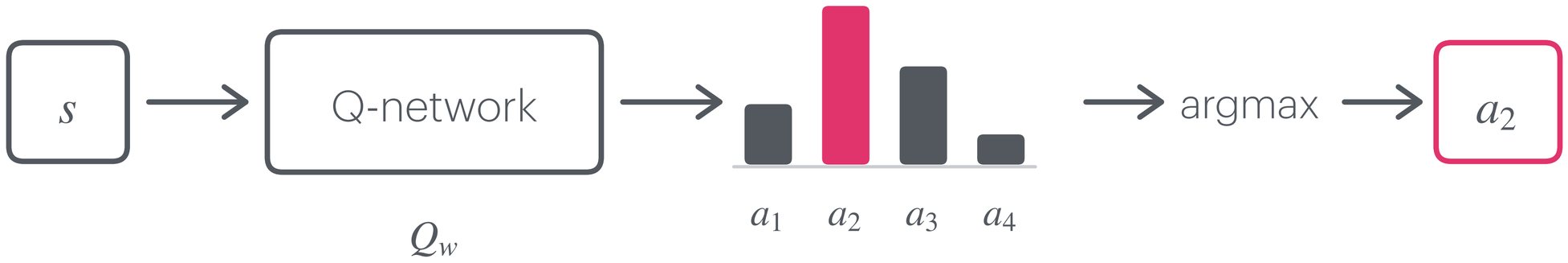
What can go wrong?

So we use a network to approximate it



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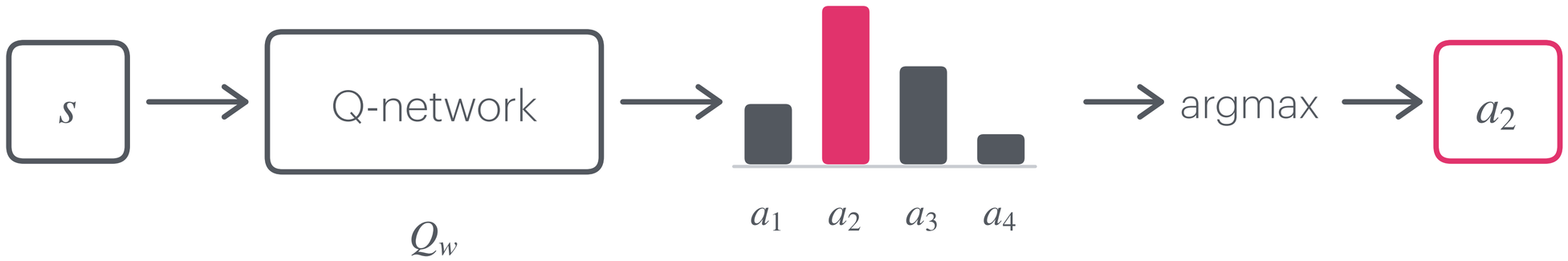
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These are all action-value methods (learn values and pick the best one)

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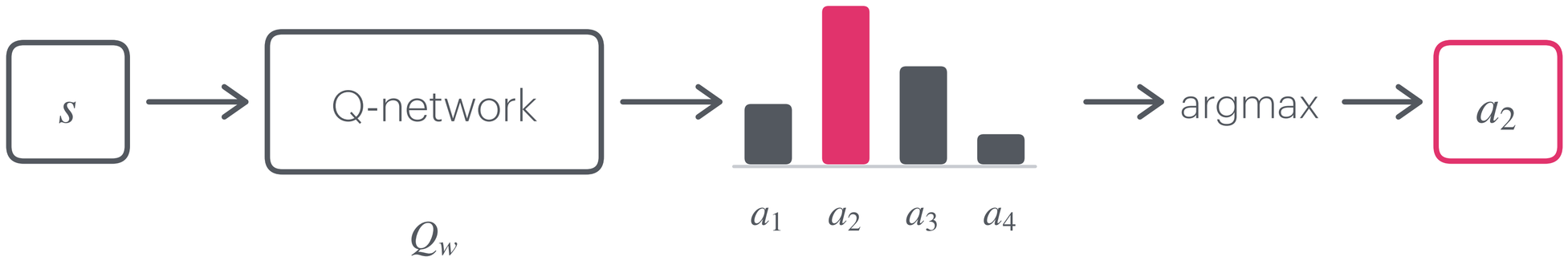


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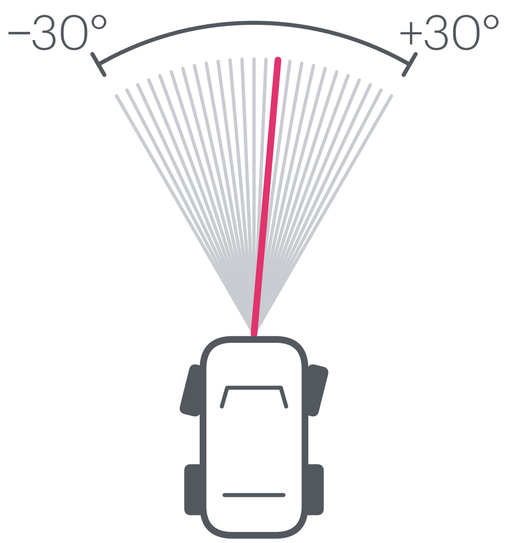


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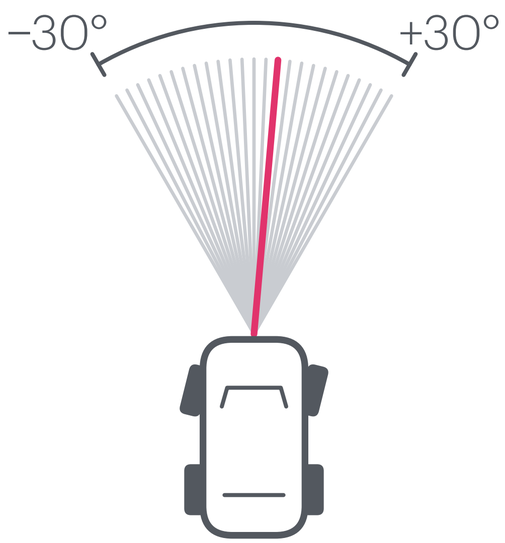
If the values are a bit off, the actions are too

Steering a car



The steering angle can be any value $a \in [-30°, 30°]$

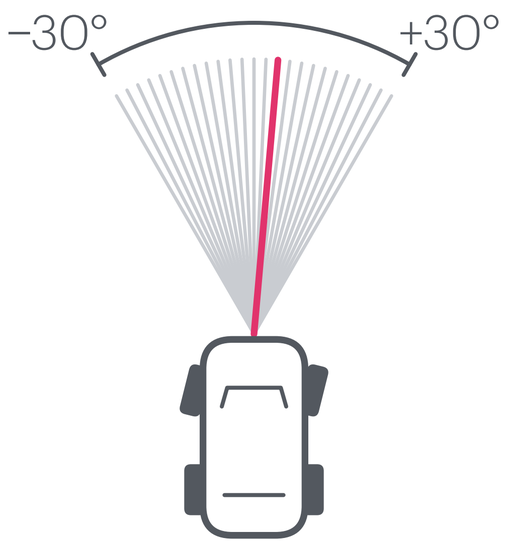
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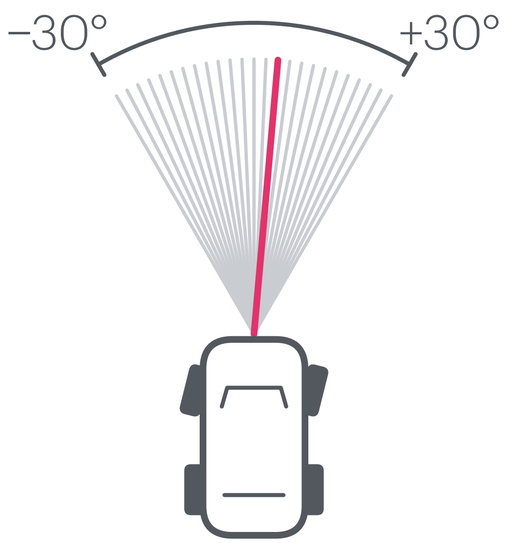


$$a^*(s) = \underset{a \in [-30°, 30°]}{\text{argmax}}\ Q_w(s, a)$$

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How do we take an argmax?

Learning the policy itself

Instead, learn the policy itself

Learning the policy itself

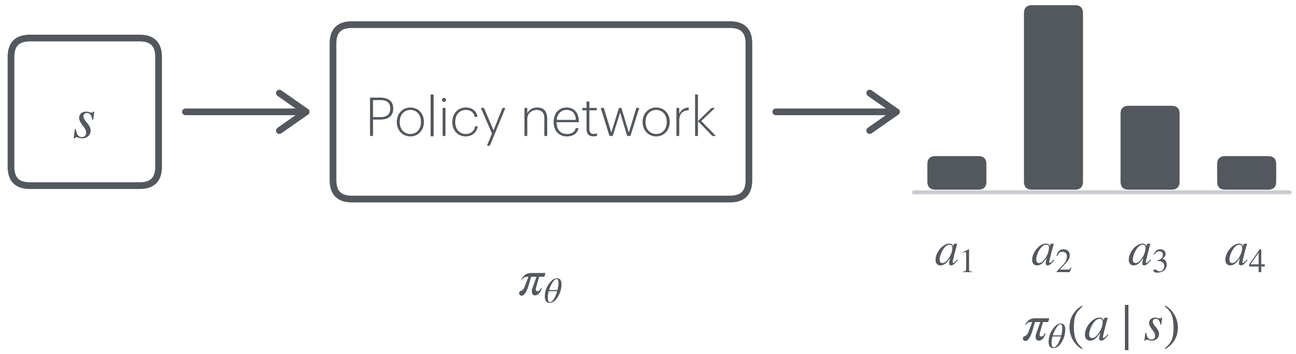
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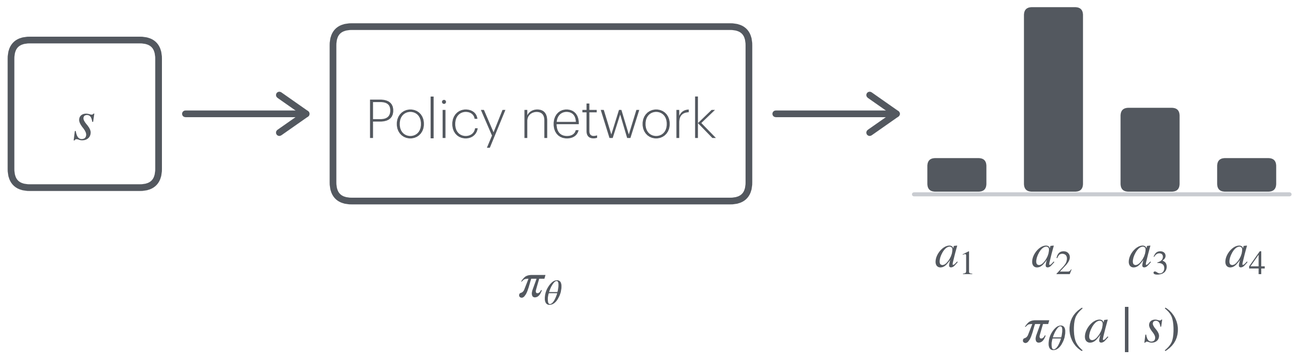


(sums to 1)

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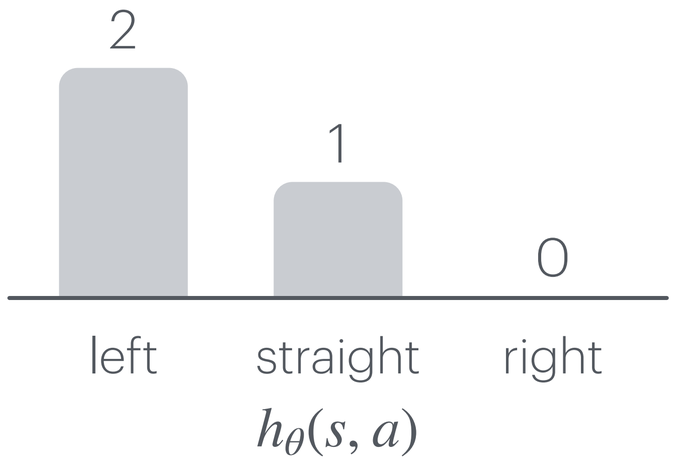


A probability distribution over actions, given the state

$$\pi_\theta(a \mid s)$$

The parameterisation

The network gives a score $h_\theta(s, a)$ for each action



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Always positive, and sums to one

Which policy is best?

Run the policy and record what happens



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$$\tau = (S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T)$$

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$$R(\tau) = \sum_{t=0}^{T-1} R_{t+1}$$

$$J(\theta) = \mathbb{E}_{\tau \sim p_\theta}\big[R(\tau)\big]$$

Which policy is best?

The best policy has the highest expected return

$$\max_{\theta} J(\theta)$$

But how do we optimise?

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$$\nabla_\theta J(\theta) = \ ?$$

REINFORCE

The log-derivative trick

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This is an expectation, so we can estimate it by sampling

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$$p_\theta(\tau) = \rho_0(s_0) \prod_{t=0}^{T-1} \pi_\theta(a_t \mid s_t)\, p(s_{t+1}, r_{t+1} \mid s_t, a_t)$$

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Putting it together

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim p_\theta}\left[R(\tau) \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t) \right]$$

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Average over episodes, not time steps

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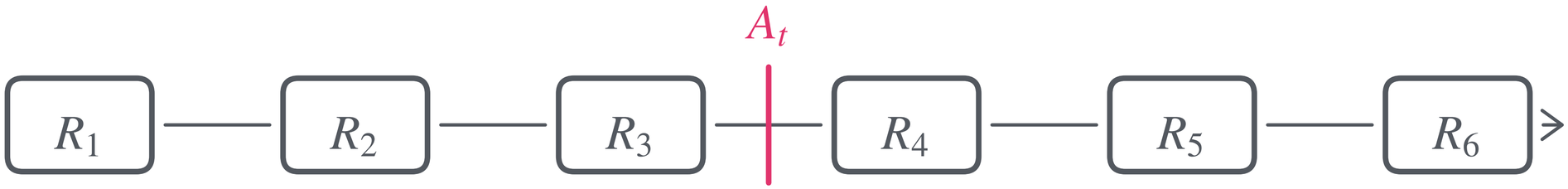
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Average over episodes, not time steps

$$\theta \leftarrow \theta + \alpha \hat{g}$$

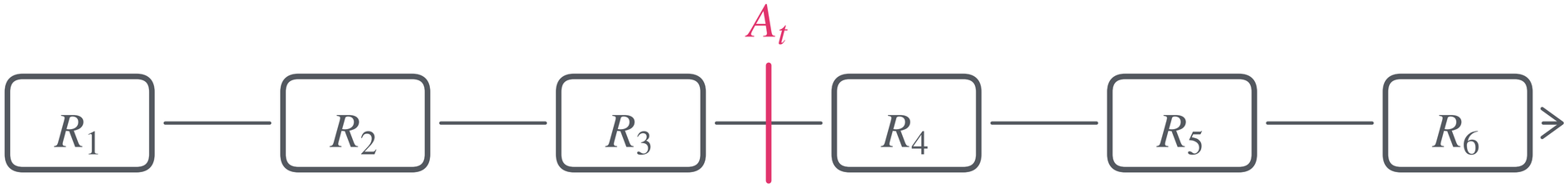
Gradient ascent, with step size α

Buttt, something looks fishy



$$R(\tau)\,\nabla_\theta \log \pi_\theta(A_t \mid S_t)$$

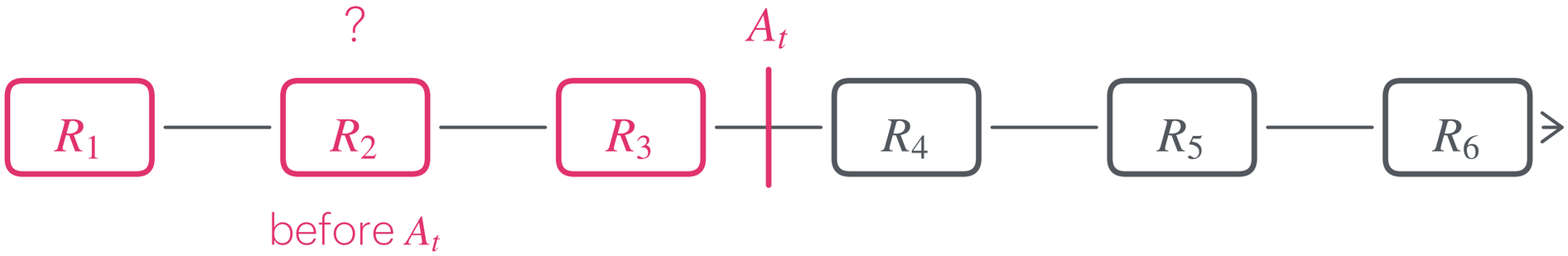
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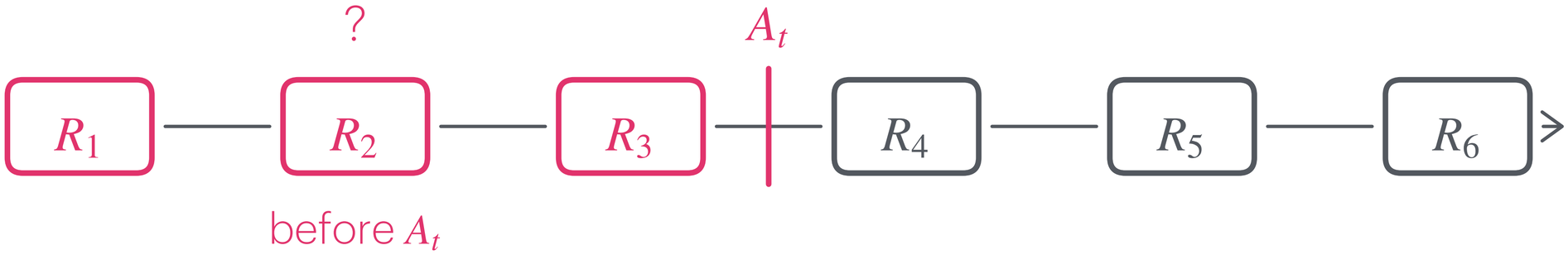


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Why does A_t get credit for earlier rewards?

Is this just noise, or is it wrong?

The policy gradient theorem

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Episodic, no discounting, fixed start state s_0

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$$\nabla_\theta J(\theta) \propto \mathbb{E}_{\substack{S \sim \mu_\theta \\ A|S \sim \pi_\theta(\cdot|S)}} [Q^{\pi_\theta}(S, A) \nabla_\theta \log \pi_\theta(A \mid S)]$$

What we need to show

$$\nabla = \nabla_\theta, \quad \pi = \pi_\theta, \quad V = V^\pi, \quad Q = Q^\pi$$

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Start with the right-hand side, written as a sum.

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Hence, it is enough to show

$$\nabla J(\theta) \propto \sum_s \mu_\theta(s) \sum_a Q(s,a)\, \nabla\pi(a|s)$$

Starting from the value

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See that

$$\nabla J(\theta) = \nabla V(s_0) = \nabla \sum_a \pi(a|s_0)\, Q(s_0, a)$$

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Using the product rule

$$= \sum_a [Q(s_0, a)\, \nabla \pi(a|s_0) + \pi(a|s_0)\, \nabla Q(s_0, a)]$$

The action value

Now look at $Q(s, a)$. By definition,

$$Q(s, a) = \sum_{s', r} p(s', r|s, a) \left[r + V(s') \right]$$

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Hence,

$$\nabla Q(s, a) = \sum_{s'} p(s'|s, a) \nabla V(s')$$

Unrolling

Substituting $\nabla Q(s, a)$ into the product rule,

$$\nabla J(\theta) = \sum_{a} Q(s_0, a)\, \nabla \pi(a|s_0) + \sum_{a} \pi(a|s_0) \sum_{s'} p(s'|s_0, a)\, \nabla V(s')$$

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call this $h(s_0)$

Unrolling, again and again

Nothing is special about s_0, so for every state s,

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Substituting again and again

$$\nabla J(\theta) = h(s_0) + \sum_{s_1} p(s_1|s_0)\, h(s_1) + \sum_{s_1} \sum_{s_2} p(s_1|s_0)\, p(s_2|s_1)\, h(s_2) + \cdots$$

Collapsing the sums

Summing out s_1

$$\sum_{s_1} p(s_1|s_0)\, p(s_2|s_1) = \quad \Pr_\pi(S_2 = s_2 \,|\, S_0 = s_0)$$

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In general, summing out the middle states

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So

$$\nabla J(\theta) = \sum_{k=0}^{\infty} \sum_{s} \Pr_\pi(S_k = s \,|\, S_0 = s_0)\, h(s)$$

Counting the visits

Swap the order of the sums

$$\nabla J(\theta) = \sum_{s} \left[\sum_{k=0}^{\infty} \Pr_{\pi}(S_k = s \mid S_0 = s_0) \right] h(s)$$

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a positive constant

Back to what we needed

Unfolding $h(s) = \sum_a Q(s,a)\nabla\pi(a|s)$,

$$\nabla J(\theta) \propto \sum_s \mu_\theta(s) \sum_a Q(s,a)\, \nabla\pi(a|s)$$

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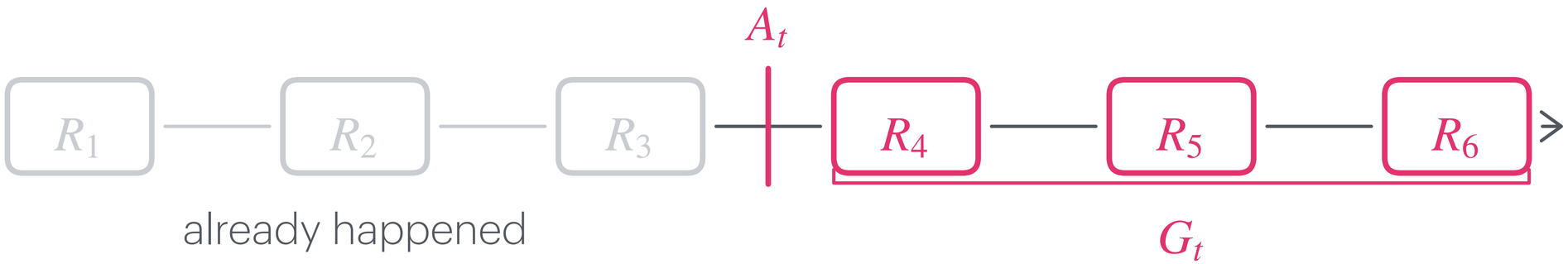
This is exactly the sum we needed to show.

Hence, the theorem

$$\nabla_\theta J(\theta) \propto \mathbb{E}_{\substack{S\sim\mu_\theta \\ A|S\sim\pi_\theta(\cdot|S)}} [Q^{\pi_\theta}(S,A)\, \nabla_\theta \log \pi_\theta(A \mid S)]$$

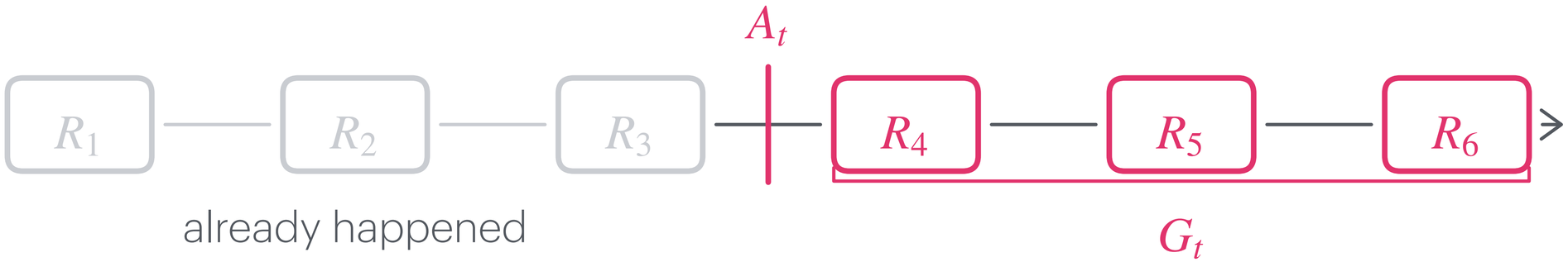
■

So, what about the past?



$$G_t = \sum_{k=t}^{T-1} R_{k+1}$$

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$$Q^{\pi_\theta}(S_t, A_t) = \mathbb{E}\left[G_t \mid S_t, A_t\right]$$

The theorem only needs rewards from now on

Replacing Q with the return

Start from the theorem

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The inner expectation disappears (tower property)

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REINFORCE

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Baselines: Reducing Noise

The goal of a baseline

Find a new estimator of the gradient:

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Same number of samples: a more accurate gradient estimate

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More reliable updates can make learning faster and more stable

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Why are we allowed to subtract it?

Because it adds zero

Given the state, b(s) is just a number

$$\mathbb{E}_{A\sim\pi(\cdot|s)}\left[b(s)\,\nabla \log \pi(A \mid s)\right] = b(s)\sum_{a}\pi(a \mid s)\,\frac{\nabla\pi(a \mid s)}{\pi(a \mid s)}$$

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Same expected gradient

A good baseline lowers the variance

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Initialize θ, $\alpha > 0$, and an action-independent baseline $b(s)$

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Actor–Critic

Where we are

REINFORCE with a baseline updates

$$\theta \leftarrow \theta + \alpha \Big(\quad G_t \quad - \quad b(S_t) \quad \Big) \nabla_\theta \log \pi_\theta(A_t|S_t)$$

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$$\theta \leftarrow \theta + \alpha \Big(\boxed{G_t} - b(S_t) \Big) \nabla_\theta \log \pi_\theta(A_t|S_t)$$

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Can we avoid waiting until the end?

Remember TD?

The return sums every future reward

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or after n steps

$$G_t^{(n)} = R_{t+1} + \cdots + R_{t+n} + V_w(S_{t+n})$$

Estimate Q, subtract V

With $b = V$, $G_t - V(S_t)$ estimates

$$Q(S_t, A_t) - V(S_t)$$

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We keep learning these estimates of Q and V on the go.

Towards PPO



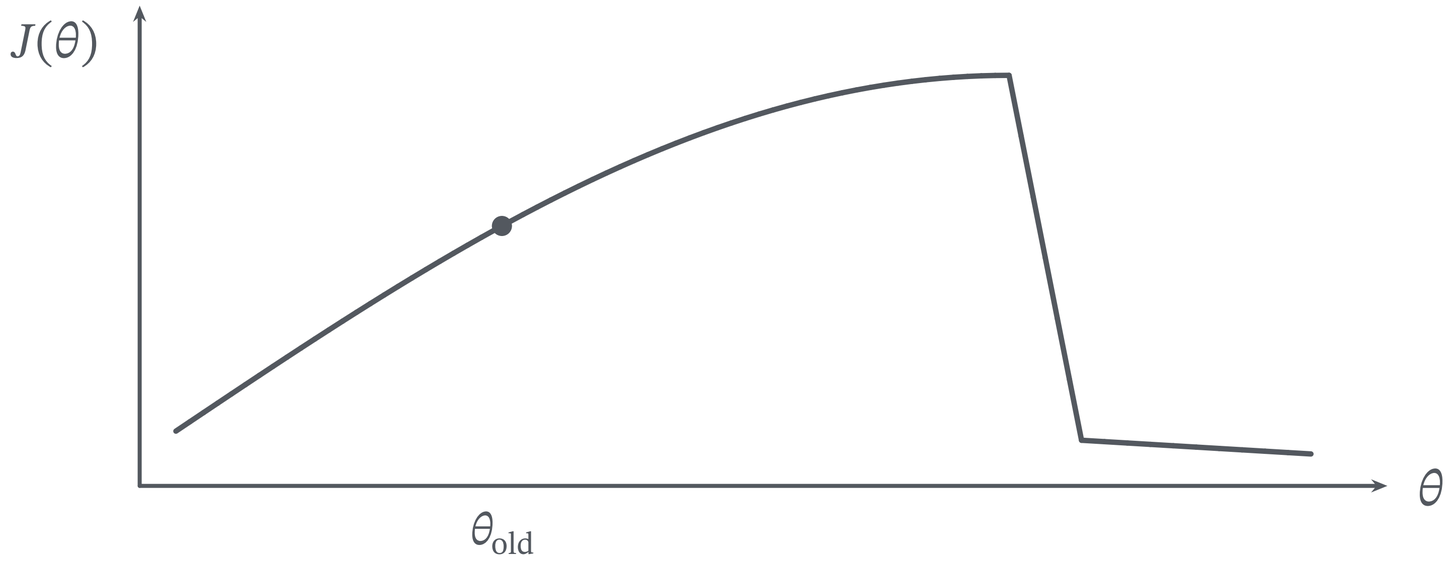
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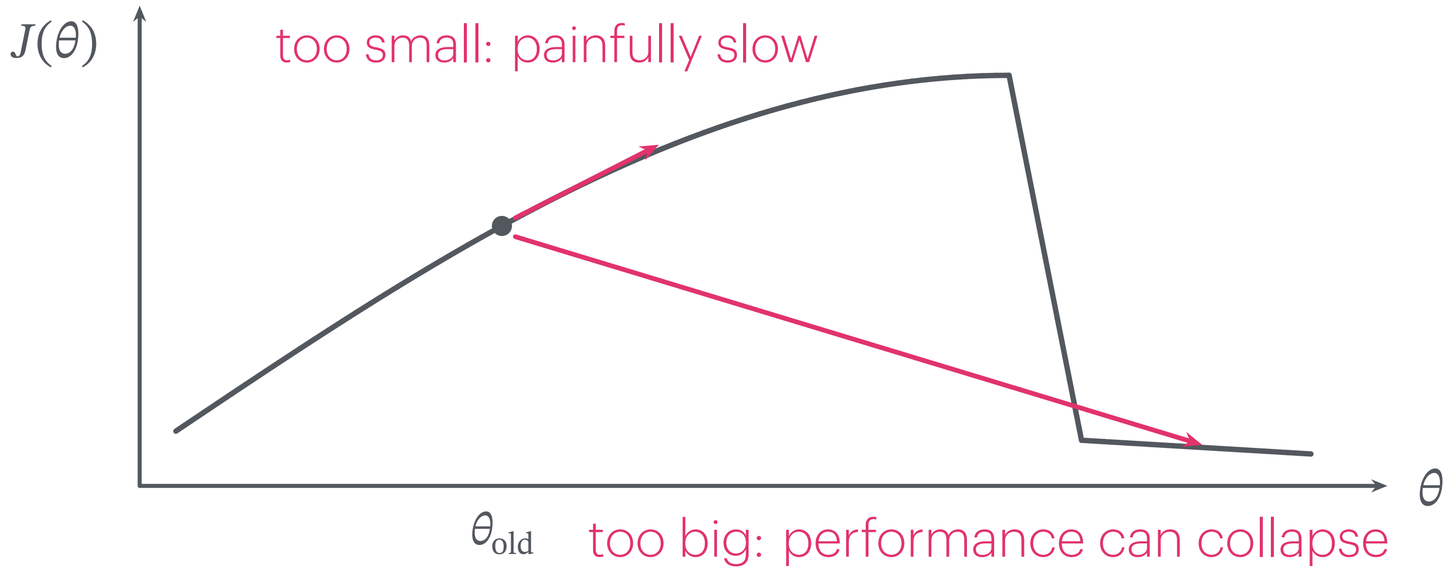
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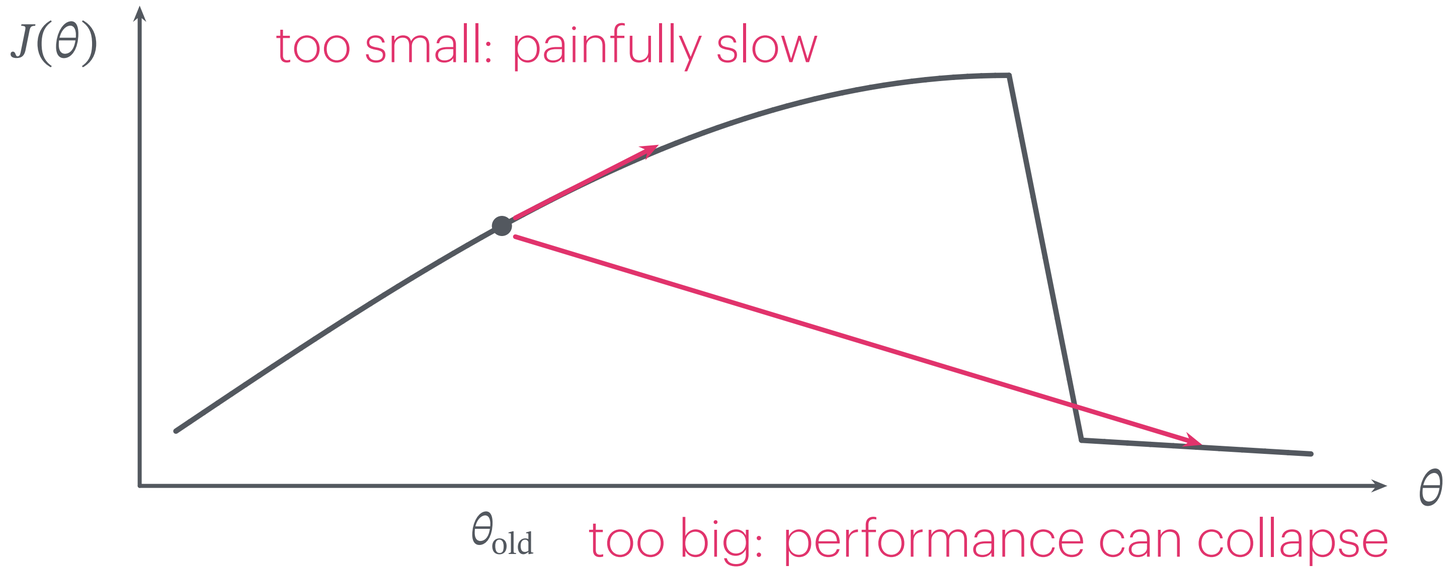
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Two problems with what we have

Each batch gives one gradient step, then we throw it away

The step size is hard to choose



Can we take a bigger step, safely, reusing the same data?

How much better is the new policy?

Suppose we are at π. How much better is a new π'?

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From here on, we discount:

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Performance difference lemma:

$$J(\pi') - J(\pi) = \mathbb{E}_{\tau \sim \pi'}\Big[\sum_{t} \gamma^t A^\pi(S_t, A_t)\Big]$$

Proof

$$\mathbb{E}_{\tau \sim \pi'}\Big[\sum_t \gamma^t A^{\pi}(S_t, A_t)\Big]$$

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the V terms telescope

$$= J(\pi') - J(\pi) \quad \blacksquare$$

since $V^\pi(s_0) = J(\pi)$

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Given π, we want to choose which π' to move to:

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needs data from π', which we haven't run

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But...

As an expectation over states

Define the discounted state distribution

$$d^{\pi}(s) = (1-\gamma)\sum_{k\geq 0}\gamma^{k}\,\mathrm{Pr}_{\pi}(S_k = s \mid S_0 = s_0)$$

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states from $d^{\pi'}$ and actions from π', but our data comes from π

Making it tractable

Fix the actions: importance sampling

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Replace it with d^{π}!

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Making it tractable

Fix the actions: importance sampling

$$\mathbb{E}_{s\sim d^{\pi'},\, a\sim\pi'(\cdot|s)}\left[A^{\pi}(s,a)\right] = \mathbb{E}_{s\sim d^{\pi'},\, a\sim\pi(\cdot|s)}\left[\frac{\pi'(a|s)}{\pi(a|s)}A^{\pi}(s,a)\right]$$

importance ratio

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Valid when $\mathrm{KL}\left(\pi'(\cdot|s)\,\|\,\pi(\cdot|s)\right)$ is small: then d^{π} and $d^{\pi'}$ are close

The surrogate objective

Finally, we have

$$L_{\pi}(\pi^{'}) = \mathbb{E}_{s\sim d^{\pi},\, a\sim\pi}\left[\frac{\pi^{'}(a|s)}{\pi(a|s)}A^{\pi}(s,a)\right]$$

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Trustworthy only while π' stays close to π.

Trust region policy optimisation (TRPO)

Take the best step that stays close

$$\max_{\theta} \; L_{\theta_{\text{old}}}(\theta) \quad \text{subject to} \quad \mathbb{E}_{s\sim d^{\pi_{\theta_{\text{old}}}}}\Big[\mathrm{KL}\big(\pi_{\theta}(\cdot|s) \,\|\, \pi_{\theta_{\text{old}}}(\cdot|s)\big)\Big] \leq \delta$$

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This needs very sophisticated, complex methods to solve.

Can we somehow use gradient descent ideas instead?

PPO: clip the ratio

The probability ratio

$$r_t(\theta) = \frac{\pi_\theta(A_t|S_t)}{\pi_{\theta_{\text{old}}}(A_t|S_t)}$$

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Keeping every ratio near 1 keeps π_θ close to $\pi_{\theta_{\text{old}}}$: a cheap trust region.

Good action: $\hat{A} > 0$

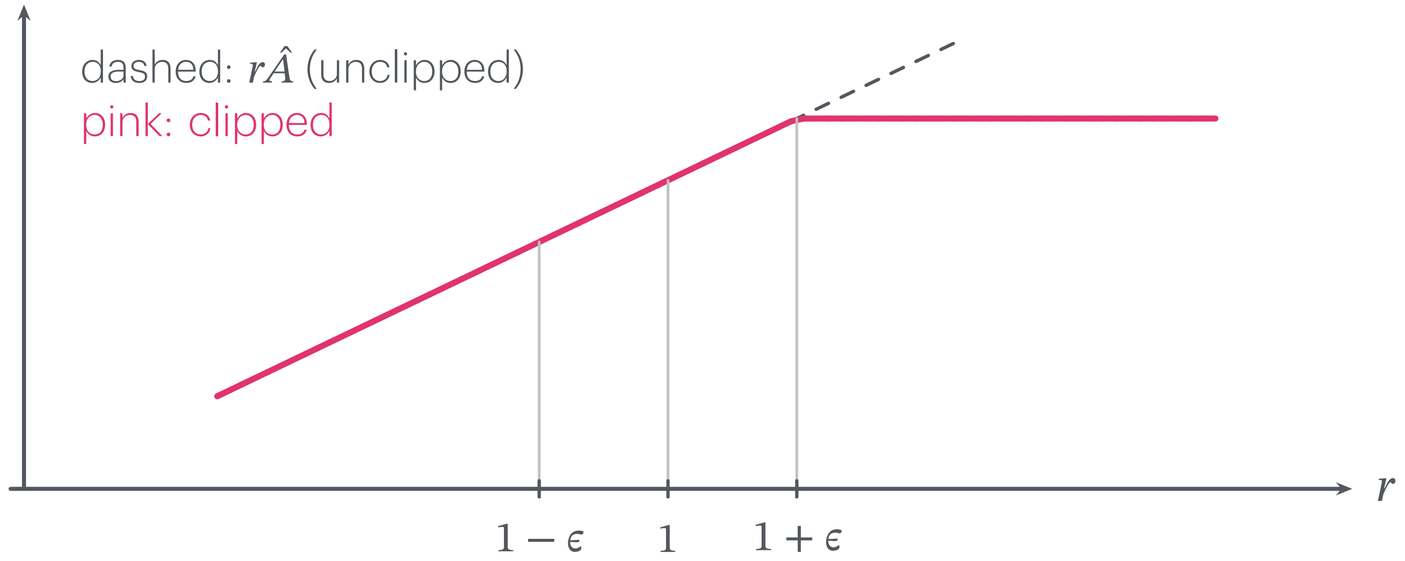
If $\hat{A}_t > 0$:

$$\min\left(r_t \hat{A}_t,\ \text{clip}(r_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\right) = \hat{A}_t \cdot \min(r_t, 1+\epsilon)$$

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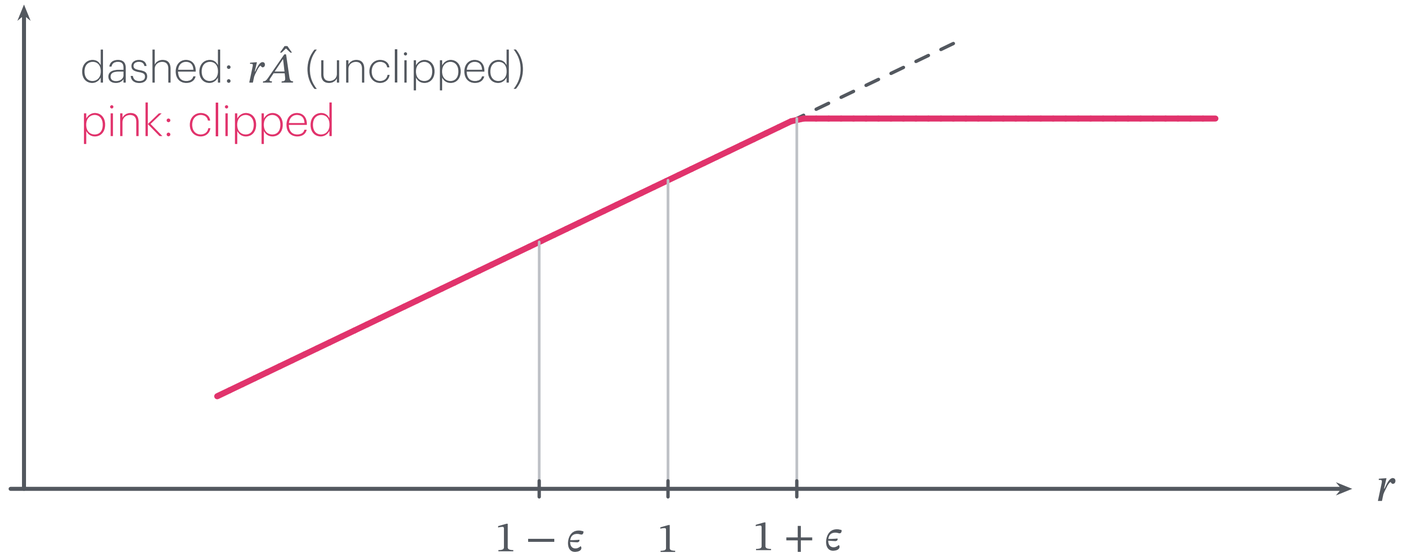
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No reward for going beyond $1+\epsilon$.

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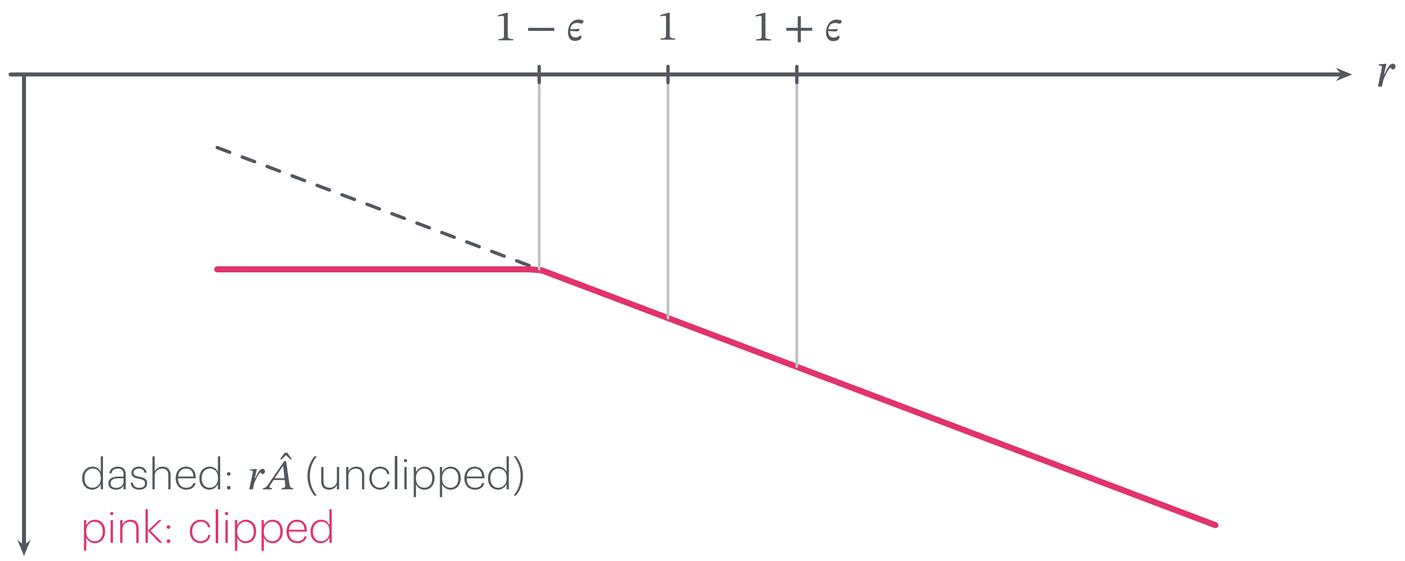
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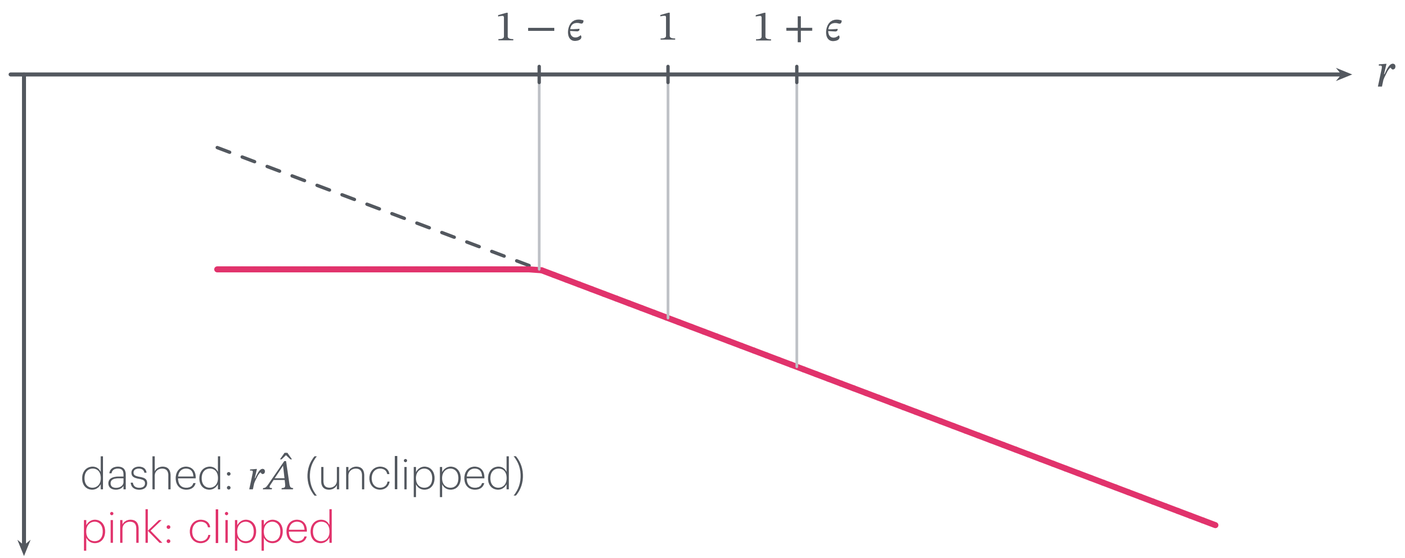
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No reward for going below $1-\epsilon$.

PPO

Initialise θ (policy) and w (critic); choose ϵ (e.g. 0.2)

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updates many times on each batch, while staying close to $\pi_{\theta_{\text{old}}}$